SQL Intermediate MCQs 75 minutes

Question - 1
MySQL: Null Value Comparison

Choose the expressions that are not correct.
Select all that apply.

- SELECT * FROM categories WHERE id <> NULL
- SELECT * FROM categories WHERE id != NULL
- SELECT * FROM categories WHERE id IS NULL
- SELECT * FROM categories WHERE id IS NOT NULL

Question - 2
MySQL: Group By Clause

Select the expression that causes a MySQL error.
Select all that apply.

- SELECT AVG(id) FROM categories
- SELECT AVG(id) FROM categories GROUP BY type
- SELECT 'average', AVG(id) FROM categories GROUP BY 1
- None of the above, all expressions are correct

Question - 3
MySQL: Nested Comments

Select the expressions that cause a MySQL error.

- SELECT * FROM /* # */ categories
- SELECT * FROM /* /* # */ categories
- SELECT * FROM /* # */ */ categories
- SELECT * FROM /* /* # */ */ categories

Question - 4
MySQL: Ampersand in Select Clause

Select the expression that causes a MySQL error.

- SELECT level & depth FROM Categories
- SELECT level && depth FROM Categories
- SELECT level &'&&'& depth FROM Categories
- SELECT level &&'&'&& depth FROM Categories
- None of the above, all expressions are correct.

Question - 5
MySQL: Pipe In Select Clause

Select the expression that causes a MySQL error.

- SELECT level | depth FROM Categories
- SELECT level || depth FROM Categories
- SELECT level ||| depth FROM Categories
- SELECT level |'|'| depth FROM Categories

Question - 6
MySQL: String Trimming

Choose the expression that retrieves the base domain "domain.com".

- SELECT LTRIM('www.domain.com', 'www.')
- SELECT TRIM(LEADING 'www.' FROM 'www.domain.com')
- SELECT TRIM(LEFT 'www.' FROM 'www.domain.com')
- All expressions are correct

Question - 7
MySQL: Substring Extraction

Choose the expression that retrieves the base domain "domain.com".

- SELECT SUBSTRING_INDEX('my.subdomain.domain.com', '.', -2)
- SELECT SUBSTRING_INDEX('my.subdomain.domain.com', 'my.subdomain.', 1)
- SELECT SUBSTRING_INDEX('my.subdomain.domain.com', '.', 2, 2)
- All expressions are correct

Question - 8
MySQL: Substring Function

Select the MySQL expression that returns a different value from the others.

- ◯ SELECT SUBSTR('user@domain.com', 5, 1)
- ◯ SELECT SUBSTR('user@domain.com' FROM 5 FOR 1)
- ◯ SELECT SUBSTR('user@domain.com' FROM -11 FOR 1)
- ◯ None of the above, all expressions return the same value.

Question - 9
MySQL: Comparing Types

Choose the expression that returns 0.

- ◯ SELECT '0' = 0
- ◯ SELECT 0 IS FALSE
- ◯ SELECT '0' IS FALSE
- ◯ SELECT STRCMP('0', 0)

Question - 10
MySQL: Repeating Strings

Select the expression that causes a MySQL error.

- ◯ SELECT STRREPEAT(' ', 10)
- ◯ SELECT SPACE(10)
- ◯ SELECT REPEAT(' ', 10)
- ◯ None of the above, all expressions are correct.

Question - 11
MySQL: ANY and ALL Operators

Select the expression that is not correct.

- ◯ The ANY and ALL operators allow the comparison of a value in one column to a range of other values.
- ◯ The ANY operator returns the first available record as the result of a query, while the ALL operator returns a subset only if the entire query condition is true.
- ◯ The ALL operator returns TRUE if all sub-query values satisfy the condition.

- The ANY operator returns TRUE if any of the sub-query values meet the condition.

Question - 12
MySQL: AUTO_INCREMENT Attribute

Which expression is not correct?

- If the column is declared NOT NULL, it is possible to assign NULL to the column to generate sequence numbers.
- Updating an existing AUTO_INCREMENT column value resets the AUTO_INCREMENT sequence.
- The initial value for AUTO_INCREMENT is fixed and always equal to 1. It will increase by 1 for each new entry.
- When the column reaches the upper limit of the data type, the next attempt to generate a sequence number fails.

Question - 13
MySQL: DEFAULT Constraint

Select the expressions that cause a MySQL error.

- CREATE TABLE users (user VARCHAR(255) DEFAULT 'test')
- CREATE TABLE users (status VARCHAR(255) DEFAULT NULL)
- CREATE TABLE users (id VARCHAR(255) DEFAULT NOT NULL)
- CREATE TABLE users (id VARCHAR(255) DEFAULT RAND())

Question - 14
MySQL: FOREIGN KEY Constraint

Which expression is not correct?

- A column might have a foreign key reference to itself.
- MySQL supports foreign key references between one column and another within a table.
- Corresponding columns in the foreign key and the referenced key must have similar data types.
- None of the above, all expressions are correct.

Question - 15
MySQL: UNIQUE Constraint

Which expression causes a MySQL error?

- CREATE TABLE users (first_name VARCHAR(255), last_name VARCHAR(255), CONSTRAINT unique_name UNIQUE (first_name, last_name))
- CREATE TABLE users (first_name VARCHAR(255) UNIQUE, last_name VARCHAR(255) UNIQUE)

- CREATE TABLE users (first_name VARCHAR(255), last_name VARCHAR(255), UNIQUE (first_name, last_name))
- None of the above, all expressions are correct.

Question - 16
MySQL: Table Constraints

Select an expression that is not a table constraint definition.

- PRIMARY KEY
- FOREIGN KEY
- CHECK
- None of the above.

Question - 17
MySQL: Table Alteration

Select the expressions that are correct.

- The ALTER TABLE statement is used to add, remove, or change existing columns in a table.
- The ALTER TABLE statement is used to add and remove various constraints in a table.
- The ALTER TABLE statement is the preferred method to delete all data in a table.
- The ALTER TABLE statement cannot be used to rename a table.

Question - 18
MySQL: Creating a Table

Which expression causes a MySQL error?

- CREATE TABLE users (id INT PRIMARY KEY)
- CREATE TABLE IF NOT EXISTS users_backup LIKE users
- CREATE TABLE IF NOT EXISTS users_backup AS SELECT * FROM users
- None of the above, all expressions are correct.

Question - 19
MySQL: Group By Condition

Select the expressions that cause a MySQL error.
Select all that apply.

- SELECT customer_id, COUNT(*) FROM transactions GROUP BY customer_id HAVING COUNT(*) > 10

- SELECT customer_id, COUNT(*) AS transactions FROM transactions GROUP BY customer_id HAVING transactions > 10
- SELECT customer_id, COUNT(*) AS transactions FROM transactions WHERE transactions > 10 GROUP BY customer_id
- SELECT customer_id, COUNT(*) AS transactions FROM transactions GROUP BY 1 HAVING transactions > 10

Question - 20
MySQL: Group Field

Select the expression that causes a MySQL error.
Select all that apply.

- SELECT customer_id, ANY_VALUE(amount) FROM transactions GROUP BY customer_id
- SELECT customer_id, RAND(amount) FROM transactions GROUP BY customer_id
- SELECT customer_id, SUBSTRING_INDEX(GROUP_CONCAT(amount ORDER BY RAND()), ',', 1) FROM transactions GROUP BY customer_id
- none of the above, all expressions are correct

Question - 21
MySQL: Group By Having

Select the expression(s) that will cause a MySQL error.
Select all that apply.

- SELECT customer_id, COUNT(*) FROM transactions GROUP BY customer_id HAVING COUNT(*) > 10
- SELECT customer_id, COUNT(*) AS transactions FROM transactions GROUP BY customer_id HAVING transactions > 10
- SELECT customer_id, COUNT(*) FROM transactions GROUP BY customer_id HAVING COUNT(customer_id) > 10
- SELECT customer_id, COUNT(*) FROM transactions GROUP BY 1 HAVING COUNT(*) > 10
- none of the above, all expressions are correct

Question - 22
MySQL: Group By Order

Which expression causes a MySQL error?
Select all that apply.

- SELECT is_active, COUNT(*) FROM transactions GROUP BY is_active ORDER BY 2
- SELECT is_active, COUNT(*) FROM transactions GROUP BY is_active ORDER BY COUNT(*)
- SELECT is_active, COUNT(*) AS transactions FROM transactions GROUP BY is_active ORDER BY transactions
- none of the above, all expressions are correct

Question - 23
MySQL: Group Variations

Select any expression(s) with the correct syntax.
Select all that apply.

- SELECT customer_id, MAX(is_active) FROM transactions GROUP BY customer_id HAVING MAX(is_active)
- SELECT customer_id, is_active FROM transactions GROUP BY customer_id HAVING MAX(is_active)
- SELECT customer_id, DISTINCT is_active FROM transactions GROUP BY customer_id HAVING MAX(is_active)
- SELECT customer_id, MAX(is_active) AS is_active FROM transactions GROUP BY customer_id HAVING is_active

Question - 24
MySQL: Distinct Select Multiple Tables

Select any expression(s) with the correct syntax.
Select all that apply.

- SELECT DISTINCT id FROM customers, customers
- SELECT a.id, DISTINCT b.id FROM customers a, customers b
- SELECT DISTINCT a.id, DISTINCT b.id FROM customers a, customers b
- SELECT DISTINCT a.id, b.id FROM customers a, customers b

Question - 25
MySQL: Substring Function

Select the expression(s) with the correct syntax.

- SELECT SUBSTRING(status, 2, 5) FROM customers
- SELECT SUBSTRING(status FROM 2 FOR 5) FROM customers
- SELECT SUBSTRING(status FROM 2 TO 5) FROM customers
- SELECT SUBSTRING(status WITH 2 TO 5) FROM customers

Question - 26
MySQL: Select Multiple Tables

Select the expressions that cause a MySQL error.
Select all that apply.

- SELECT a.id, b.id FROM customers a, customers b
- SELECT id, id FROM customers, customers

- SELECT id FROM customers, customers
- SELECT a.id, id FROM customers a, customers

Question - 27
SQL Statements

Which of the following statement(s) are **NOT** correct:

- The PRIMARY KEY must be unique and not null for each table.
- The DROP command is used to remove the table definition and its contents whereas the TRUNCATE command is used to delete all the rows from the table.
- DELETE command is a DDL command whereas DROP is a DML command.
- ACID properties in databases refer to Atomicity, Complexity, Isolation, and Duplicacy.